

TechMart Electronics

Official Technical Support Knowledge Base

Login Issues | Password Reset | Installation
Guide | Troubleshooting | API Errors | System Requirements

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Classification: Public — for use by customers, support agents, and AI assistants

This document is a fictional company knowledge base created for educational use in an AI/RAG (Retrieval-Augmented Generation) capstone project. TechMart Electronics is not a real company.

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Note: Page numbers above are approximate and correspond to the structure of this knowledge base. This document is intended to be chunked and embedded into a vector database (e.g., FAISS or ChromaDB) for use by the Technical Support Agent of the Multi-Agent AI Customer Support Assistant.

0. About This Technical Support Guide

This document is the companion technical knowledge base to TechMart Electronics' billing knowledge base, and covers account access, device setup, troubleshooting, developer API integration, and system requirements across TechMart's website, mobile app, and smart home ecosystem. It is intended to ground the Technical Support Agent of TechMart's AI-powered customer support assistant, and may also be referenced by the Product Agent when a question spans both product capability and technical setup.

Products and Platforms Covered

- **TechMart Website** — www.techmart-electronics.example, desktop and mobile browser experience.
- **TechMart App** — native iOS and Android application for shopping, order tracking, and smart home device control.
- **TechMart SmartHome Platform** — the HomeHub controller and companion app ecosystem covering cameras, doorbells, locks, thermostats, and lighting.
- **TechMart Developer API** — REST API used by internal TechMart applications and by approved third-party integration partners.
- **TechMart Vision Smart TVs** — TechMart's smart TV operating system and associated streaming apps.

How to Use This Document

Each chapter is organized so that a support agent (human or AI) can locate the specific symptom a customer describes and follow a defined resolution path. Chapters 1 and 2 (Login Issues, Password Reset) cover account access; Chapter 3 (Installation Guide) covers first-time device and app setup; Chapter 4 (Troubleshooting) covers ongoing operational issues after successful setup; Chapter 5 (API Errors) is intended primarily for developers integrating with TechMart's API, but is also useful when a customer-facing error message includes an API error code; and Chapter 6 (System Requirements) documents minimum and recommended specifications for every supported platform.

1. Login Issues

This chapter covers issues customers may encounter signing in to their TechMart account across the website, mobile app, and smart home app, including incorrect credentials, account lockouts, two-factor authentication (2FA) problems, social sign-in issues, and session/device management.

1.1 Standard Login Methods

- **Email + Password** — the default sign-in method for all TechMart accounts.
- **Sign in with Apple** — available on iOS app and Safari browser.
- **Sign in with Google** — available on all platforms.
- **Business SSO (SAML 2.0)** — available for verified TechMart Business Accounts using an approved identity provider (Okta, Azure AD, Google Workspace).

1.2 "Incorrect Email or Password" Error

This is the most common login error and can result from a genuinely mistyped password, a forgotten password, or an account originally created using a social sign-in method (Apple or Google) being entered with the email-and-password form instead. Because TechMart does not disclose which factor was wrong for security reasons, the error message is intentionally generic. Recommended troubleshooting order: (1) verify the email address is typed correctly with no extra spaces, (2) confirm Caps Lock is off, (3) try the "Forgot Password" flow described in Chapter 2, and (4) if the customer originally signed up via Apple or Google, direct them to use that same button rather than entering a password.

1.3 Account Lockouts

After 5 consecutive failed login attempts within 15 minutes, TechMart temporarily locks the account for 30 minutes as a brute-force protection measure. During a lockout, even a correct password will return a "too many attempts" message rather than allowing sign-in, which is expected behavior. Customers can bypass the lockout timer immediately by successfully completing the password reset flow (Chapter 2), which also resets the failed-attempt counter. Repeated lockouts across multiple days from different IP addresses may indicate a credential-stuffing attempt against the account and are automatically flagged for the Trust & Fraud Prevention team to review.

1.4 Two-Factor Authentication (2FA) Issues

1.4.1 Not Receiving the 2FA Code

TechMart 2FA codes are delivered via SMS text message or a time-based one-time password (TOTP) authenticator app, depending on which method the customer enabled under Account Settings > Security > Two-Factor Authentication. SMS codes can be delayed by carrier network congestion, particularly for prepaid or international carriers; TOTP codes generated by an authenticator app (such as a standard

authenticator app) do not depend on network delivery at all and are the more reliable option. If an SMS code does not arrive within 2 minutes, the customer can request a new code using the "Resend Code" link, which invalidates the previous code.

1.4.2 TOTP Code "Invalid" Errors

The most common cause of a TOTP code being rejected as invalid is clock drift between the customer's phone and TechMart's authentication server; TOTP codes are time-based and only valid for a 30-second window. Advising the customer to ensure their phone's date and time are set to update automatically resolves the large majority of these cases. If codes still fail after confirming automatic time sync, the authenticator entry may need to be removed and re-added by scanning a fresh QR code from Account Settings > Security.

1.4.3 Lost Access to 2FA Device

Customers who lose access to their 2FA device (phone lost, stolen, or replaced without transferring the authenticator app) can use one of the 10 backup codes generated when 2FA was first enabled, viewable and downloadable at that time from Account Settings > Security > Backup Codes. If no backup codes were saved, the customer must go through TechMart's identity verification process with a human support agent, which requires a government-issued photo ID and typically takes 1-2 business days, since this is a security-sensitive account recovery path that cannot be automated.

1.5 Social Sign-In Issues

When a customer signs in with Apple or Google, TechMart matches the account using the email address provided by Apple or Google. If a customer previously created a TechMart account with the same email using a password, the systems are automatically linked, and either sign-in method will work going forward. If Apple's "Hide My Email" relay feature was used at signup, the customer must continue using Sign in with Apple to access that account, since TechMart never learns the customer's real email address in that scenario and cannot offer a standard password-based fallback.

1.6 Business SSO Login Issues

Business Account users signing in via SAML SSO who receive an "SSO configuration error" should be directed to their organization's IT administrator first, since this error indicates a misconfiguration on the identity provider side (incorrect Assertion Consumer Service URL, expired signing certificate, or an attribute mapping issue) rather than a TechMart account problem. TechMart's Business Accounts team can provide the organization's current SAML metadata XML file upon request from an authorized administrator to help diagnose the configuration.

1.7 Session and Device Management

Customers can view all devices currently signed in to their account under Account Settings > Security > Active Sessions, and remotely sign out any device from that screen. Sessions automatically expire after 90 days of inactivity on the website, and after 180 days on the mobile app (to reduce the need for frequent re-authentication on a personal device). Signing out of one device does not affect other active sessions.

1.8 Login Issue Quick Reference

Symptom	Likely Cause	Recommended Action
"Incorrect email or password"	Typo, forgotten password, or wrong sign-in method used	Verify typed credentials; try Forgot Password; check if account is active
"Too many attempts, try again later"	Account temporarily locked after 5 failed attempts	Wait 30 minutes, or complete password reset to unlock immediately
2FA code never arrives (SMS)	Carrier delay or incorrect phone number on file	Resend code; verify phone number; switch to TOTP authentication
TOTP code says invalid	Clock drift between phone and server	Enable automatic date/time on the device; re-add authentication
Locked out with no 2FA device	Lost phone, no backup codes saved	Escalate to identity verification with a human agent
"This email is already in use" when trying to sign up	Account already exists, possibly via social sign-in	Sign in with Apple/Google, or use Forgot Password
SSO configuration error (Business Accounts)	Identity provider misconfiguration	Refer to the organization's IT administrator; provide TechMart SSO IDP configuration
Logged out unexpectedly	Session expiration or manual sign-out from another device	Sign in again; review Active Sessions for unrecognized devices

1.9 Login Issues on Shared or Family Devices

On shared household devices, such as a tablet used by multiple family members, TechMart recommends signing out fully rather than relying on "Remember Me" to avoid one family member's saved payment methods or order history being visible to another. TechMart Plus Family accounts (Chapter 3 of the Billing Knowledge Base) are the recommended way to give household members their own sign-in while still sharing membership benefits, rather than sharing a single login.

1.10 Region and Storefront Mismatch

A customer whose account was originally created on the Canadian storefront but who later moves to the United States (or vice versa) may see a "This account is registered to a different region" message when attempting to sign in from the other country's version of the site or app. This is expected behavior, since pricing, currency, and tax handling differ by storefront; the customer should contact Billing Support to discuss migrating their account, rather than creating a duplicate account, which would fragment their order and warranty history.

1.11 Browser Cookie and Cache Issues

Persistent login failures on the website that do not match any of the causes above are frequently resolved by clearing the browser's cookies and cache for techmart-electronics.example specifically, or by testing in a private/incognito window to rule out a corrupted local session cookie. This is particularly common after a recent browser update or after a browser extension related to privacy or ad-blocking has been installed or updated.

1.12 In-Store Loyalty Login (Register Lookup)

Customers checking out in-store can link their TechMart account at the register by providing their phone number or email instead of signing in on a device, which allows the purchase to sync to their online order

history and count toward TechMart Plus benefits such as member pricing. If the lookup fails to find an account, the associate can manually search by name and postal code as a fallback before concluding no account exists.

2. Password Reset

This chapter explains the password reset process for TechMart accounts across all platforms, including standard self-service reset, reset link troubleshooting, password requirements, and the assisted account-recovery path for customers who cannot complete self-service reset.

2.1 Standard Password Reset Flow

- 1 From the sign-in screen, select "Forgot Password."
- 2 Enter the email address associated with the TechMart account.
- 3 TechMart sends a password reset email containing a secure, one-time link, typically arriving within 1–2 minutes.
- 4 Click the link, which opens a Set New Password page; the link is valid for 30 minutes from the time it was sent.
- 5 Enter and confirm a new password that meets TechMart's password requirements (Section 2.3).
- 6 Upon success, the customer is automatically signed in and all other active sessions are signed out for security.

2.2 Reset Link Troubleshooting

Symptom	Likely Cause	Recommended Action
Reset email never arrives	Filtered to spam, or email typed incorrectly	Check spam/promotions folder; verify the email matches
"This link has expired"	More than 30 minutes since the email was sent	Request a new reset email; each link is single-use
"This link is invalid"	Link was copied incompletely, or a newer reset email was requested	Use the most recently received reset email; avoid manual
Reset succeeds but old password still works	Previously active session on another device	Sign out manually or wait for automatic sign out

2.3 Password Requirements

- Minimum 10 characters.
- At least one uppercase letter, one lowercase letter, and one number.
- At least one special character (e.g., ! @ # \$ % ^ & *).
- Cannot match any of the account's 5 most recently used passwords.
- Cannot contain the customer's email address or name.

2.4 Changing a Password While Signed In

Customers who are already signed in and simply wish to change their password (rather than having forgotten it) can do so under Account Settings > Security > Change Password, which requires entering the current password once as a confirmation step before the new password is accepted. This path does not send an email link, since the customer is already authenticated.

2.5 Password Reset for Business SSO Accounts

Business Account users who sign in exclusively via SAML SSO do not have a TechMart-managed password at all; their credentials are managed entirely by their organization's identity provider. These users should be directed to their organization's own password reset process (e.g., their company's Okta or Azure AD portal) rather than TechMart's Forgot Password flow, which will not apply to their account.

2.6 Assisted Account Recovery

If a customer no longer has access to the email address on file and therefore cannot receive a reset link, they must go through TechMart's assisted account recovery process with a human support agent. This requires verifying identity through at least two of the following: a government-issued photo ID, the last four digits of a payment method previously used on the account, and answers to account security questions if previously set. Assisted recovery typically takes 1-2 business days and cannot be completed by the AI Billing or Technical Support Agent directly, since it requires a manual identity verification step performed by an authorized human agent.

2.7 Security Notes

TechMart never sends a customer's actual password by email, phone, or chat, since passwords are stored using a one-way cryptographic hash and are not retrievable even by TechMart engineers. Any communication claiming to contain a customer's actual current password should be treated as a phishing attempt and reported through Account Settings > Security > Report Suspicious Activity.

2.8 Password Reset on the Mobile App vs. Website

The password reset flow is identical whether initiated from the mobile app or the website, since both rely on the same reset-link email; however, tapping a reset link on a phone that has the TechMart App installed will open the Set New Password screen directly inside the app via a deep link, while the same link opens in the mobile browser if the app is not installed. Customers occasionally report the link "doing nothing" when it silently opens the app in the background; checking if the app was already open and simply switched to the Set New Password screen usually resolves this confusion.

2.9 Password Managers and Autofill

TechMart's sign-in and password reset forms are built to standard HTML autocomplete attributes and are compatible with major password managers. Occasionally a password manager will autofill an outdated saved password after a reset; customers experiencing a login failure immediately after a successful reset should double-check that their password manager has saved the new password rather than continuing to offer the old one.

2.10 Reset Attempt Rate Limiting

To prevent abuse, TechMart limits password reset email requests to 5 per hour per account. Customers who repeatedly request reset emails in a short period (for example, while troubleshooting a delivery issue) may see a "Please wait before requesting another reset email" message; this is a security control and resolves automatically after the hourly window passes.

2.11 Password Reset Quick Reference

Situation	Correct Path
Forgot password, has access to account email	Standard Forgot Password flow (Section 2.1)
Wants to change password while signed in	Account Settings > Security > Change Password
Business SSO account	Organization's own identity provider password reset
No access to account email and no phone on file	Assisted account recovery with a human agent
Suspects account was compromised	Reset password immediately, then enable 2FA if not already active, then contact support

3. Installation Guide

This chapter provides step-by-step installation and setup instructions for the TechMart app and for TechMart's most commonly purchased connected devices.

3.1 Installing the TechMart App

3.1.1 iOS

- 1 Open the App Store and search "TechMart Electronics."
- 2 Tap Get, then authenticate the download with Face ID, Touch ID, or an Apple ID password.
- 3 Once installed, open the app and choose Sign In or Create Account.
- 4 Allow notifications when prompted, to receive order and delivery updates.

3.1.2 Android

- 1 Open the Google Play Store and search "TechMart Electronics."
- 2 Tap Install.
- 3 Once installed, open the app and choose Sign In or Create Account.
- 4 Grant requested permissions (notifications, and Bluetooth/location if setting up smart home devices).

3.1.3 Common App Installation Issues

Symptom	Likely Cause	Recommended Action
App not found in store	Regional store restriction or search term mismatch	Confirm the device's app store region is set to United States
"Insufficient storage" during install	Device storage below app's space requirement (see table)	Free up at least 250MB and retry
App crashes immediately on open	Outdated OS version, or a corrupted install	Update device OS to the minimum supported version; re-install
Stuck on splash screen	Poor network connectivity during first launch	Connect to stable Wi-Fi and relaunch the app

3.2 Setting Up TechMart SmartHome Devices

3.2.1 HomeHub Central Controller (Set Up First)

- 1 Plug in the HomeHub near your Wi-Fi router for initial setup.
- 2 Open the TechMart App and go to Smart Home > Add Device > HomeHub.
- 3 Scan the QR code on the bottom of the HomeHub using your phone's camera within the app.
- 4 Select your home Wi-Fi network and enter the password when prompted.

- 5 Wait for the HomeHub's status light to turn solid blue, indicating a successful connection.
- 6 Name the HomeHub and assign it to a room (e.g., "Living Room") for organization purposes.

The HomeHub should be set up before any other smart home devices, since cameras, locks, thermostats, and lighting all pair through the HomeHub rather than connecting to Wi-Fi individually in most installations, which improves reliability and reduces Wi-Fi network congestion.

3.2.2 GuardCam Security Camera / Video Doorbell

- 1 In the TechMart App, go to Smart Home > Add Device > Camera or Doorbell.
- 2 Hold the device's pairing button for 5 seconds until the LED blinks amber.
- 3 The app will detect the device automatically if the HomeHub is already set up; follow the on-screen pairing steps.
- 4 Mount the camera or doorbell using the included mounting bracket and screws, ensuring a clear view of the intended area.
- 5 For video doorbells, connect the two low-voltage wires to your existing doorbell wiring, or use battery power if no existing wiring is present.

3.2.3 SmartLock Pro Deadbolt

- 1 Remove the existing deadbolt following the included removal guide.
- 2 Install the SmartLock Pro exterior and interior assemblies using the provided screws.
- 3 Insert 4 AA batteries into the interior assembly.
- 4 In the TechMart App, go to Smart Home > Add Device > Smart Lock and follow the pairing prompt.
- 5 Test the lock by locking and unlocking it from the app before relying on it as your primary lock.

3.2.4 ClimateSense Smart Thermostat

- 1 Turn off power to your HVAC system at the breaker before beginning installation.
- 2 Remove the old thermostat and label each wire according to its terminal letter.
- 3 Mount the ClimateSense base plate and connect wires to the matching labeled terminals.
- 4 Restore power and follow the in-app setup wizard under Smart Home > Add Device > Thermostat.
- 5 Complete the guided HVAC system type selection (e.g., central air, heat pump) so ClimateSense can calibrate correctly.

3.3 Setting Up a TechMart Vision Smart TV

- 1 Connect the TV to power and turn it on; the setup wizard starts automatically on first boot.
- 2 Select your language and connect to Wi-Fi, or connect an Ethernet cable for a wired connection.

- 3 Sign in with your TechMart account, or create one, to sync purchase history and enable the TechMart Streaming Bundle if applicable.
- 4 Allow the TV to complete its initial software update, which can take 5-15 minutes depending on connection speed.
- 5 Complete the picture calibration step, which optimizes brightness and color for your room's lighting.

3.4 Installing a Printer (PrintFast Series)

- 1 Unbox the printer and remove all shipping tape and protective packaging from the print carriage.
- 2 Install the included starter ink or toner cartridges as shown in the quick-start guide.
- 3 Connect the printer to Wi-Fi using the printer's built-in touchscreen or control panel menu.
- 4 Install the TechMart PrintFast driver on your computer from techmart-electronics.example/drivers, or use the AirPrint/Mopria mobile printing standard for phones and tablets without installing a driver.
- 5 Print a test page from the printer's control panel to confirm setup before printing from a device.

3.5 Installing a Router or Mesh System (MeshLink)

- 1 Connect the primary MeshLink unit to your modem using the included Ethernet cable.
- 2 Download the TechMart App and go to Smart Home > Add Device > MeshLink Router.
- 3 Follow the guided setup to name your network and set a Wi-Fi password.
- 4 For multi-unit mesh systems, place satellite units within range of the primary unit (generally one room away) and add each one following the same in-app flow.
- 5 Run the built-in speed test from the app after setup to confirm expected performance.

3.6 Driver and Firmware Downloads

Drivers for printers, webcams, and other PC-connected TechMart peripherals are available at techmart-electronics.example/drivers by searching the product's SKU or model name. Firmware updates for smart home devices, routers, and smart TVs are delivered automatically over Wi-Fi and do not require a manual download in normal operation; manual firmware files are only provided for advanced troubleshooting at the direction of Technical Support.

3.7 Installation Support Escalation

Customers who are not comfortable performing electrical installations themselves (such as the ClimateSense thermostat or SmartLock Pro, which involve existing home wiring or hardware) can book a paid TechMart Professional Installation appointment through the app under Smart Home > Professional Installation, available in most metro areas served by TechMart's retail locations.

3.8 Setting Up GlowSet Smart Lighting

- 1 Screw in the GlowSet smart bulb like a normal bulb, or mount the light strip/floodlight per its included hardware.
- 2 Power on the fixture; the bulb will flash briefly to indicate it is in pairing mode.
- 3 In the TechMart App, go to Smart Home > Add Device > Lighting.
- 4 Select the light from the discovered devices list and assign it to a room.
- 5 Optionally group multiple lights together under Smart Home > Groups for simultaneous control.

3.9 Setting Up SweepBot Robot Vacuum

- 1 Charge the SweepBot fully on its charging dock before first use.
- 2 In the TechMart App, go to Smart Home > Add Device > Robot Vacuum and follow the pairing prompt.
- 3 Run the initial mapping cycle, during which SweepBot scans the home to build a floor map.
- 4 Once mapping is complete, label rooms in the app to enable room-specific cleaning commands.
- 5 For the Vacuum & Mop Combo model, fill the onboard water tank before starting a mopping cycle.

3.10 Setting Up ImmerseOne VR Headset

- 1 Charge the headset fully before first use.
- 2 Download the companion ImmerseOne app (separate from the TechMart App) to a compatible phone or PC.
- 3 Put on the headset and follow the guided room-scale boundary setup, walking the edges of your play area as prompted.
- 4 Pair both controllers by holding their pairing buttons while inside the headset's setup menu.
- 5 Sign in with your TechMart account inside the headset to sync any purchased content.

3.11 Pre-Installation Checklist

- Confirm your home Wi-Fi network name and password are on hand before starting any smart home setup.
- Update your phone's TechMart App to the latest version before beginning setup.
- For any device involving existing home wiring (thermostats, video doorbells, locks), turn off the relevant breaker before disconnecting anything.
- Keep original packaging until setup is confirmed successful, in case a return or exchange is needed.
- Register your product warranty automatically by ensuring you're signed in to your TechMart account during setup; this links the purchase and setup date for warranty tracking without a separate registration step.

4. Troubleshooting

This chapter addresses issues that arise after a device or app has already been successfully set up, organized by product category.

4.1 General First Steps (Apply to Nearly All Devices)

- 1 Restart the device (power off fully, wait 10 seconds, power back on).
- 2 Restart the TechMart App or close and reopen the browser tab.
- 3 Confirm the device and phone/computer are on the same Wi-Fi network, especially for smart home devices.
- 4 Check for a pending firmware or app update under Settings > About > Check for Updates.
- 5 If multiple devices are affected at once, check techmart-electronics.example/status for a known service outage.

4.2 App and Website Issues

Symptom	Likely Cause	Recommended Action
App stuck loading / spinning indefinitely	Poor connectivity or a temporary server-side issue	Check techmart-electronics.example/status; restart app;
Blank white screen on website	Browser extension conflict or outdated browser	Try an incognito/private window; update the browser to t
Cart items disappear between sessions	Not signed in, or cookies cleared	Sign in before shopping so the cart is saved to the accou
Push notifications not arriving	Notifications disabled at the OS level	Check phone Settings > Notifications > TechMart A
App shows outdated order status	Local cache not yet refreshed	Pull down to refresh on the Orders screen

4.3 Smart Home Connectivity Issues

Symptom	Likely Cause	Recommended Action
Device shows "Offline" in app	Device lost Wi-Fi connection or was power-cycled	Restart the device, reconnecting Wi-Fi network is up; re-pair if it
Camera video is choppy or low resolution	Insufficient upload bandwidth or the camera is too far from the router	Reduce the video resolution in-app; consider a Wi-Fi exte
Smart lock unresponsive from app but works on a physical key	Bluetooth/Wi-Fi bridge issue between lock and the central hub	Moved the HomeHub closer to the lock, or add a range-ext
Motion alerts not triggering	Motion sensitivity set too low, or the sensor zone is not properly configured	Adjust sensitivity and detection zones under the device's
Thermostat not reaching set temperature	HVAC system mismatch during setup, or a wiring issue	Re-run the HVAC system type wizard; if unresolved, this
Devices randomly go offline overnight	Router's Wi-Fi power-saving mode disconnected devices	Disable Wi-Fi power-saving/DTIM interval settings on the

4.4 Smart TV Issues

Symptom	Likely Cause	Recommended Action
No picture but TV powers on	Incorrect input source selected	Cycle through inputs using the Source/Input button
Streaming apps buffer constantly	Insufficient internet speed for the selected video quality	Lower streaming quality in-app; test speed via Settings > Network
Remote not responding	Depleted batteries or Bluetooth pairing lost	Replace batteries; re-pair remote via Settings > Remote
TV won't complete software update	Update interrupted by power loss or weak Wi-Fi	Ensure stable power and Wi-Fi during the update; retry from the TV's home screen
Audio out of sync with video	Processing delay from an external soundbar	Enable Lip Sync / Audio Delay adjustment in the TV's sound settings

4.5 Computing Accessory and Peripheral Issues

Symptom	Likely Cause	Recommended Action
Wireless mouse/keyboard lagging or disconnecting	Low battery, USB interference, or too far from receiver	Replace batteries; move the USB receiver closer or use a USB extension cable
External monitor not detected	Loose cable or unsupported display mode	Reseat the cable; try a different port; confirm resolution/refresh rate
USB-C hub not charging laptop	Hub's power delivery wattage insufficient for the laptop	Use a higher-wattage USB-C hub, or charge the laptop directly
Mechanical keyboard keys not registering	Debris under keycaps or outdated firmware	Clean gently with compressed air; check for a firmware update

4.6 Printer Issues

Symptom	Likely Cause	Recommended Action
Print jobs stuck in queue	Printer offline or driver communication issue	Restart printer and computer; remove and re-add the printer
Faded or streaky prints	Low ink/toner or clogged print head	Run the built-in print head cleaning cycle from the printer's control panel
Wi-Fi printing stopped working after router update	Printer still connected to the old network	Re-run Wi-Fi setup on the printer's control panel with the new network
Paper jams repeatedly	Incorrect paper type/size loaded, or worn feed rollers	Confirm paper matches tray settings; clean feed rollers with a lint-free cloth

4.7 Audio Device Issues

Symptom	Likely Cause	Recommended Action
Bluetooth headphones won't pair	Device still paired to another source, or Bluetooth module disabled	Forget the device on all previously paired sources, then re-pair
One earbud not producing sound	Earbuds out of sync or low battery on one side	Return both earbuds to the case for 10 seconds to force sync
Speaker cuts out intermittently	Bluetooth interference or out-of-range	Move closer to the source device; reduce nearby Wi-Fi/Bluetooth devices
Microphone not detected by computer	Wrong input device selected in OS sound settings	Select the correct input device in the computer's sound settings

4.8 Battery and Power Issues (Wearables, Micromobility)

Devices that fail to hold a charge significantly below their rated battery life after the first few months of use, despite being charged with the original or an approved replacement charger, should be evaluated for a

warranty battery claim rather than continued troubleshooting, since battery degradation beyond expected wear may indicate a defective battery cell covered under the standard manufacturer warranty described in the Billing Knowledge Base.

4.15 On-Device Error Code Glossary

Several TechMart hardware products display short on-device error codes on an LED indicator pattern or small screen. The table below decodes the most common codes reported to Technical Support.

Device	Code / Indicator	Meaning	Action
HomeHub	Blinking red, 3x pattern	Wi-Fi connection lost	Check router; re-run Wi-Fi setup if it does not reconnect
HomeHub	Solid red	Internal fault	Power cycle; contact Technical Support if it persists
GuardCam	Blinking amber, slow	Pairing mode active	Normal during setup; complete pairing in the app
GuardCam	Blinking amber, fast	Firmware update in progress	Do not unplug; wait for solid status light
SmartLock Pro	3 short beeps + red flash	Low battery	Replace all 4 AA batteries as soon as possible
SmartLock Pro	Continuous beeping	Bolt jammed / obstruction	Check door alignment; do not force the mechanism
ClimateSense	"E1" on display	Wiring/terminal mismatch	Re-run the HVAC wizard; verify wire labels match terminal
ClimateSense	"E3" on display	Sensor fault	Contact Technical Support; may require a warranty replacement
MeshLink Router	Solid amber	No internet connection detected	Check modem connection and cabling
MeshLink Router	Blinking blue	Satellite unit out of range	Move the satellite closer to the primary unit or another satellite
PrintFast Printer	Blinking ink icon	Low or empty cartridge	Replace the indicated cartridge
PrintFast Printer	Blinking paper icon	Paper jam or empty tray	Clear the paper path or refill the tray
SweepBot	3 beeps + red light	Stuck or wheel obstructed	Relocate the unit to open floor space and restart the cycle
RollEasy Scooter/Bike	Blinking battery icon	Critically low battery	Charge immediately; avoid deep discharge for battery longevity

4.16 Factory Reset Instructions

Device	Factory Reset Method
HomeHub	Hold the reset button on the bottom for 15 seconds until the light flashes white
GuardCam / Doorbell	Hold the pairing button for 20 seconds until the LED flashes rapidly amber
SmartLock Pro	Remove the interior cover and hold the reset pinhole button for 10 seconds
ClimateSense	Hold the physical button on the unit for 10 seconds, or reset via Settings > Advanced > Factory Reset
MeshLink Router	Hold the recessed reset button for 10 seconds until all lights blink simultaneously
SweepBot	Settings > Advanced > Factory Reset in-app, or hold the physical power button for 10 seconds

Factory resetting a device removes it from the account entirely and erases its local configuration; customers should only factory reset a device as a last-resort troubleshooting step or before selling or gifting the device

to someone else, since it will need to be fully re-paired afterward.

- The issue involves potential electrical safety concerns (e.g., burning smell, visible damage, overheating).
- The customer has already completed all standard troubleshooting steps without resolution.
- The issue affects a professionally installed device (thermostat wiring, video doorbell wiring).
- The customer requests a hands-on remote screen-share troubleshooting session.
- A firmware or software bug is suspected that may affect multiple customers, which should be logged for the engineering team.

4.10 Robot Vacuum Issues

Symptom	Likely Cause	Recommended Action
SweepBot won't return to dock	Dock moved, obstructed, or map outdated	Ensure a clear 3-foot radius around the dock; re-run map
Suction seems weak	Full dustbin or clogged brush roll	Empty the dustbin and clear hair/debris from the brush r
Mopping leaves streaks	Dirty mop pad or low water flow	Rinse or replace the mop pad; check the water tank is p

4.11 VR Headset Issues

Symptom	Likely Cause	Recommended Action
Tracking drifts or jitters	Insufficient lighting in the play area	Increase room lighting; avoid reflective surfaces like mirr
Controller not detected	Low battery or lost pairing	Replace/charge batteries; re-pair from the headset's setu
Display looks blurry	Headset not properly adjusted to the user's eyes	Adjust the lens spacing (IPD) slider until the image is sha

4.12 E-Bike and Scooter Issues (RollEasy)

Symptom	Likely Cause	Recommended Action
Won't power on	Battery fully discharged or not seated properly	Charge for at least 30 minutes and confirm the battery is
Reduced range compared to spec	Cold weather, incline riding, or aging battery	Range naturally drops in cold weather and hilly terrain; c
App won't connect to the scooter/bike	Bluetooth not enabled or too far from the device	Enable Bluetooth on the phone; stay within about 30 fee

4.13 Firmware Update Failures

If a smart home device, router, or TV fails to complete a firmware update (commonly shown as stuck at a percentage, or reverting after a restart), the most common causes are an interrupted power supply during the update or an unstable Wi-Fi connection. Devices should remain powered and connected throughout an update, which for most TechMart devices takes between 3 and 15 minutes. If a device becomes unresponsive after a failed update, most TechMart smart home devices support a factory-reset recovery

mode (typically a 10-15 second hold of the physical reset button) that allows the device to be re-paired and will retry the firmware update automatically during re-setup.

4.14 Diagnosing Whole-Home Wi-Fi Problems

When a customer reports that many unrelated devices are having connectivity issues simultaneously (smart home devices, the TV, and computers all struggling at once), the root cause is very likely the home Wi-Fi network or internet connection itself rather than any individual TechMart product. Recommended diagnostic steps: confirm the modem and router both show normal status lights, restart the modem and router in that order with a 30-second gap, and run a speed test from a device connected directly to the router. If speeds are far below the customer's internet plan even after a restart, the issue should be directed to their internet service provider rather than TechMart.

4.17 Known Issues and Workarounds

The following are current known issues that TechMart's engineering team is aware of and actively working to resolve. Support agents should reference these directly rather than running full troubleshooting steps when a customer's symptom matches one below, since standard troubleshooting will not resolve an issue that is already confirmed as a known bug.

GuardCam night vision occasionally shows a green tint

Affects a subset of GuardCam units manufactured before March 2025. A firmware fix is in testing; in the meantime, slightly increasing the camera's IR sensitivity in the app reduces the effect.

SweepBot Vacuum & Mop Combo may pause mid-clean near reflective flooring

The unit's obstacle sensor can misread highly reflective tile as a drop-off edge. Workaround: enable "Reduced Cliff Sensitivity" under Settings > Advanced for affected rooms.

MeshLink satellite units occasionally show as offline after a primary unit firmware update

A satellite may need to be manually power-cycled once after the primary unit completes an update. This is expected to be resolved in a future firmware release that better coordinates update timing across mesh units.

TechMart App on certain older Android tablets shows an incorrect order total on the Orders screen

A display-only rounding issue affecting a small number of older Android tablet models; the invoice and actual charge are always correct. A permanent fix is planned; in the meantime, customers can confirm the exact total from Account > Order History > Invoice.

PulseFit Smartwatch SE step count may under-report during cycling workouts

The SE model's motion algorithm is optimized for walking/running; cycling activities should be logged using the dedicated Cycling workout mode rather than relying on automatic step detection.

4.18 When to Escalate to Human Technical Support

- The issue involves potential electrical safety concerns (e.g., burning smell, visible damage, overheating).
- The customer has already completed all standard troubleshooting steps without resolution.
- The issue affects a professionally installed device (thermostat wiring, video doorbell wiring).
- The customer requests a hands-on remote screen-share troubleshooting session.
- A firmware or software bug is suspected that may affect multiple customers, which should be logged for the engineering team.

This chapter documents the TechMart Developer API, used internally by TechMart's own applications and by approved third-party integration partners (such as smart home automation platforms) to access order status, subscription status, and SmartHome device data on a customer's behalf, with the customer's authorization.

5.1 Authentication

The TechMart Developer API uses OAuth 2.0 with the authorization code grant for third-party integration partners, and API keys for TechMart's own internal applications. Access tokens are valid for 1 hour; refresh tokens are valid for 90 days of inactivity. All requests must be made over HTTPS; plain HTTP requests are rejected outright.

```
POST /oauth/token
Content-Type: application/json

{
  "grant_type": "authorization_code",
  "code": "AUTH_CODE_HERE",
  "client_id": "YOUR_CLIENT_ID",
  "client_secret": "YOUR_CLIENT_SECRET",
  "redirect_uri": "https://yourapp.example/callback"
}
```

5.2 Standard Error Response Format

All TechMart API errors are returned as JSON with a consistent structure:

```
{
  "error": {
    "code": "INVALID_TOKEN",
    "message": "The access token provided is expired or invalid.",
    "status": 401,
    "request_id": "req_8f2a1c9d3b"
  }
}
```

The request_id should always be included when reporting an API issue to TechMart Developer Support, since it allows the engineering team to locate the exact request in server logs without needing full request/response payloads, which may contain sensitive customer data.

5.3 HTTP Status Code Reference

Status Code	Meaning	Typical Cause
400 Bad Request	Malformed request syntax	Missing required field, invalid JSON, or invalid parameter type
401 Unauthorized	Missing or invalid authentication	Expired access token, missing Authorization header, or bad API key
403 Forbidden	Authenticated but not permitted	Token lacks the required OAuth scope for this endpoint
404 Not Found	Resource does not exist	Invalid order ID, customer ID, or device ID
409 Conflict	Request conflicts with current resource state	Attempting to cancel an order that has already shipped
422 Unprocessable Entity	Valid syntax but semantically invalid	A field value fails business rule validation (e.g., negative quantity)
429 Too Many Requests	Rate limit exceeded	Client exceeded the allowed requests per minute (see Section 5.5)
500 Internal Server Error	Unexpected server-side failure	Should be reported to TechMart Developer Support with the request_id
503 Service Unavailable	TechMart API temporarily down	Scheduled maintenance or an active incident; check the status page

5.4 Application-Level Error Codes

Error Code	HTTP Status	Description
INVALID_TOKEN	401	The access token is expired, revoked, or malformed
MISSING_SCOPE	403	The token does not include the OAuth scope required for this endpoint
CUSTOMER_NOT_FOUND	404	No customer record matches the provided customer ID
ORDER_NOT_FOUND	404	No order matches the provided order ID
DEVICE_NOT_LINKED	404	The specified SmartHome device is not linked to this customer's account
ORDER_ALREADY_SHIPPED	409	Cannot cancel or modify an order that has entered shipment
SUBSCRIPTION_ALREADY_CANCELLED	409	Cannot cancel a subscription that is already cancelled
VALIDATION_FAILED	422	One or more fields failed validation; see the 'details' array in the response
RATE_LIMIT_EXCEEDED	429	Too many requests sent in the current rate-limit window
WEBHOOK_DELIVERY_FAILED	n/a (async)	TechMart could not deliver a webhook event after repeated retries

5.5 Rate Limits

The TechMart Developer API enforces a default rate limit of 100 requests per minute per API client, and 1,000 requests per minute for approved high-volume Business Integration partners. Every response includes rate limit headers so client applications can self-throttle proactively:

```
X-RateLimit-Limit: 100
X-RateLimit-Remaining: 42
X-RateLimit-Reset: 1735689600
```

When the limit is exceeded, the API returns HTTP 429 with a Retry-After header indicating the number of seconds to wait before the next request will be accepted. Developers are strongly encouraged to implement exponential backoff rather than retrying immediately after a 429 response, since immediate retries from many clients simultaneously can worsen load during an incident.

5.6 Webhooks

TechMart can push real-time events (order status changes, subscription renewals, device status changes) to a partner's registered webhook URL rather than requiring the partner to poll the API. TechMart retries a failed webhook delivery up to 5 times with exponential backoff over roughly 24 hours before marking the event as permanently failed and logging a `WEBHOOK_DELIVERY_FAILED` event, viewable in the Developer Portal's Webhook Logs. Every webhook payload includes an HMAC-SHA256 signature in the `X-TechMart-Signature` header, which partner applications must verify to confirm the request genuinely originated from TechMart.

5.7 Common Integration Mistakes

- Not refreshing the access token before it expires, resulting in intermittent 401 errors.
- Hardcoding endpoint URLs from documentation examples instead of using the base URL provided at app registration, which differs between the sandbox and production environments.
- Not handling 429 responses gracefully, causing cascading failures during traffic spikes.
- Failing to verify the webhook HMAC signature, which is a security requirement, not an optional step.
- Assuming order and subscription IDs are numeric; they are opaque string identifiers and should not be parsed or reformatted by client code.

5.8 Sandbox Environment

TechMart provides a full sandbox environment at `sandbox-api.techmart-electronics.example` with synthetic test data, allowing developers to test integrations without affecting real customer data or triggering real charges. Sandbox API keys are prefixed with `sk_test_` and production keys with `sk_live_`, making it easy to detect if a key was accidentally used in the wrong environment; TechMart's API rejects any request where the key environment does not match the requested base URL.

5.10 Core API Endpoints Reference

Endpoint	Method	Description
<code>/v1/customers/{id}</code>	GET	Retrieve basic customer profile information
<code>/v1/orders/{id}</code>	GET	Retrieve order status, items, and tracking information
<code>/v1/orders/{id}/cancel</code>	POST	Cancel an order prior to shipment
<code>/v1/subscriptions/{id}</code>	GET	Retrieve subscription plan, status, and next renewal date
<code>/v1/subscriptions/{id}/cancel</code>	POST	Cancel a subscription effective end of current cycle

Endpoint	Method	Description
/v1/devices/{id}	GET	Retrieve SmartHome device status and last-seen timestamp
/v1/devices/{id}/events	GET	Retrieve recent event history for a SmartHome device
/v1/webhooks	POST	Register a new webhook endpoint for event delivery

5.11 Example Request and Response

```
GET /v1/orders/TM-2026-0481932
Authorization: Bearer ACCESS_TOKEN
```

Response 200 OK:

```
{
  "order_id": "TM-2026-0481932",
  "status": "shipped",
  "carrier": "RegionalFreight",
  "tracking_number": "RF1029384756",
  "estimated_delivery": "2026-03-18"
}
```

5.12 Pagination

List endpoints (such as retrieving a customer's order history) are paginated using cursor-based pagination rather than page numbers, to remain stable even as new records are added. Each response includes a `next_cursor` field; passing this value as the cursor query parameter on the next request retrieves the following page. A null `next_cursor` indicates the final page of results.

```
GET /v1/customers/cus_8821/orders?limit=20&cursor;=eyJpZCI6MTIzfQ
```

Response 200 OK:

```
{
  "data": [ ... ],
  "next_cursor": "eyJpZCI6MTQzfQ",
  "has_more": true
}
```

5.13 Idempotency

State-changing requests (such as order cancellation) accept an optional Idempotency-Key header. If the same key is sent twice, TechMart's API returns the original response rather than performing the action a second time, which protects against duplicate actions caused by network retries. Idempotency keys are honored for 24 hours after the original request.

5.14 Deprecation Policy

TechMart provides a minimum of 12 months' notice before removing any API endpoint or field, communicated through the Developer Portal changelog and, for registered API clients, by email to the account's technical contact. Deprecated fields continue to be returned alongside their replacement during

the notice period, marked with a deprecated: true flag in the API documentation.

Developer-specific issues (authentication setup, webhook configuration, unexpected error codes) should be directed to developer-support@techmart-electronics.example rather than general Billing or Technical Support, since API issues are handled by TechMart's Platform Engineering team and require the request_id and API client ID to investigate effectively.

6. System Requirements

6.1 TechMart App — Mobile

Platform	Minimum Version	Recommended	Storage Needed
iOS	iOS 16.0	iOS 17.0 or later	250 MB free space
Android	Android 10 (API 29)	Android 13 or later	250 MB free space
iPadOS	iPadOS 16.0	iPadOS 17.0 or later	300 MB free space

6.2 TechMart Website — Desktop Browsers

Browser	Minimum Version	Notes
Google Chrome	110 or later	Fully supported; recommended for best performance
Mozilla Firefox	110 or later	Fully supported
Apple Safari	16 or later	Fully supported on macOS and iPadOS
Microsoft Edge	110 or later	Fully supported (Chromium-based)
Internet Explorer	Not supported	TechMart's website will display a browser upgrade notice

6.3 HomeHub Central Controller Requirements

- Home Wi-Fi network with 2.4GHz support (5GHz optional, used for higher-bandwidth camera streaming when available).
- Minimum internet upload speed of 2 Mbps per connected camera for standard-definition recording, 5 Mbps for high-definition.
- A dedicated electrical outlet within 6 feet of the router for initial setup (the HomeHub can be relocated after setup completes).
- iOS 16+ or Android 10+ device for initial app-based setup.

6.4 TechMart Vision Smart TV Requirements

Requirement	Minimum	Recommended
Internet speed for streaming (1080p)	5 Mbps	10 Mbps or higher
Internet speed for streaming (4K)	15 Mbps	25 Mbps or higher
Wi-Fi standard	802.11n (Wi-Fi 4)	802.11ac (Wi-Fi 5) or 802.11ax (Wi-Fi 6)
HDMI input (for external devices)	HDMI 1.4	HDMI 2.1 for full 4K/120Hz gaming support

6.5 PrintFast Printer Requirements

- Wi-Fi network (2.4GHz) for wireless setup; USB connection also supported for direct computer connection.
- Windows 10 or later, or macOS 12 or later, for driver installation.
- AirPrint (iOS) or Mopria Print Service (Android) supported for driverless mobile printing.

6.6 MeshLink Router / Mesh System Requirements

Requirement	Detail
Internet connection type	Cable, fiber, or DSL modem with an available Ethernet port
Minimum recommended internet plan	100 Mbps for a single MeshLink unit; 300 Mbps+ for a 3-unit mesh system with many connected devices
Setup device	iOS 16+ or Android 10+ smartphone with the TechMart App
Power	Dedicated electrical outlet per unit; no built-in battery backup

6.7 TechMart Developer API Requirements

- TLS 1.2 or later for all API requests.
- Ability to handle JSON request and response bodies.
- A publicly reachable HTTPS endpoint if using webhooks (localhost/private IPs are not reachable by TechMart's servers).
- Registered API client credentials obtained from the TechMart Developer Portal at developers.techmart-electronics.example.

6.8 Wearables (PulseFit) Compatibility

Device	iOS Requirement	Android Requirement	Bluetooth
PulseFit Smartwatch Series 3	iOS 16+	Android 10+	Bluetooth 5.0+
PulseFit Smartwatch SE	iOS 16+	Android 10+	Bluetooth 5.0+
TrackBand Fitness Tracker	iOS 15+	Android 9+	Bluetooth 4.2+

6.10 Network Ports and Firewall Requirements

Households with advanced router configurations, custom firewalls, or corporate/campus networks (relevant primarily for Business Account SmartHome deployments) should ensure outbound access on standard HTTPS port 443 to *.techmart-electronics.example and its content delivery subdomains, and outbound UDP on the ephemeral port range for real-time camera streaming (typically ports 10000-20000, though TechMart

devices will automatically negotiate a working port via STUN/TURN relay if direct UDP is blocked, at some cost to video latency).

6.11 Storage and Memory Recommendations by Use Case

Use Case	Recommendation
Casual app use (shopping, order tracking only)	Minimum requirements are sufficient
Multiple SmartHome cameras with local app viewing	Recommended phone spec, plus a strong Wi-Fi connection
4K streaming on TechMart Vision TV	25 Mbps+ internet connection, HDMI 2.1 source devices for full quality
Large SmartHome deployment (10+ devices)	5GHz-capable router or mesh system to reduce 2.4GHz congestion
Business API integration	No specific hardware requirement; server-side compute scales with request volume

6.12 Legacy Device Support Policy

TechMart supports the two most recent major OS versions for iOS and Android at any given time, consistent with the minimum versions listed in Section 6.1. Devices running an older OS may still be able to install a previous version of the TechMart App but will not receive new feature updates and may eventually lose the ability to sign in as backend security requirements are strengthened. TechMart provides at least 90 days' notice by in-app banner before deprecating support for an OS version.

6.13 Accessibility Features

- Full VoiceOver (iOS) and TalkBack (Android) screen reader support across the TechMart App.
- Adjustable text size independent of system settings, under App Settings > Display.
- High-contrast mode available on the TechMart Vision Smart TV interface.
- Closed captioning support for all TechMart-produced setup and troubleshooting videos.

For the most reliable experience across all TechMart connected devices, TechMart recommends a home network with a router no more than 3-4 years old, a minimum internet plan of 100 Mbps for households with multiple smart home devices and streaming TVs, and keeping the TechMart App and all connected device firmware set to automatic updates, which is the default setting and can be adjusted under Settings > Updates in the app.

7. Appendix

7.1 Glossary of Technical Support Terms

2FA — Two-Factor Authentication; a second verification step beyond a password, typically an SMS or app-generated code.

API — Application Programming Interface; allows software systems to communicate with TechMart's platform programmatically.

Firmware — Low-level software embedded on a device (e.g., a smart lock or router) that controls its core functionality.

HMAC Signature — A cryptographic signature used to verify that a webhook payload genuinely came from TechMart and was not tampered with.

HomeHub — TechMart's central smart home controller that bridges other smart home devices to the TechMart App.

OAuth 2.0 — An industry-standard authorization framework used by the TechMart Developer API for secure third-party access.

Rate Limit — A cap on how many API requests a client may make in a given time window.

SSO — Single Sign-On; allows Business Account users to sign in using their organization's existing identity provider.

TOTP — Time-based One-Time Password; a 2FA code generated by an authenticator app that changes every 30 seconds.

Webhook — An automated, real-time HTTP notification TechMart sends to a partner application when an event occurs.

7.2 Technical Support Escalation Matrix

Issue Type	First Point of Contact	Escalation Path
Routine login/setup question	AI Technical Support Agent	Human Technical Support agent
Account recovery (no 2FA access)	Human Technical Support agent	Identity Verification team
Suspected security incident	Human Technical Support agent	Trust & Fraud Prevention team
Suspected multi-customer firmware bug	Technical Support agent	Engineering / Product team
Developer API issue	developer-support@techmart-electronics.example	Platform Engineering team
Electrical/safety concern with a device	Technical Support agent	Immediate escalation; advise customer to stop u

7.3 Contact Directory

Department	Phone	Email
General Technical Support	1-800-555-0188	techsupport@techmart-electronics.example
Developer Support	N/A (email/portal only)	developer-support@techmart-electronics.example
SmartHome Professional Installation	1-800-555-0144	installs@techmart-electronics.example
Trust & Fraud Prevention	1-800-555-0166	fraud@techmart-electronics.example

7.4 Sample AI Technical Support Agent Conversations

2FA Code Not Arriving

Customer: I never get my login code by text.

Technical Support Agent: Sorry about that — SMS codes can occasionally be delayed by carrier networks. I'd recommend switching to an authenticator app under Account Settings > Security > Two-Factor Authentication, which generates codes on your device without needing a text message. In the meantime, you can also request a fresh code using Resend Code, which sometimes goes through faster.

Smart Camera Offline

Customer: My GuardCam shows offline in the app but the light is on.

Technical Support Agent: Let's start with the basics — could you try unplugging the camera for 10 seconds and plugging it back in? If it's still offline after a minute or two, check that your Wi-Fi network is up by testing another device. If the network's fine but the camera still won't reconnect, we may need to re-pair it through Smart Home > Add Device in the app.

API 429 Errors

Developer: We're getting 429 errors during our nightly sync job.

Technical Support Agent: That indicates you're exceeding the rate limit — 100 requests per minute by default. I'd recommend checking the X-RateLimit-Remaining header in your responses and implementing exponential backoff when you see it approaching zero. If your integration needs a higher sustained volume, we can evaluate you for our Business Integration tier at 1,000 requests per minute.

7.5 Frequently Referenced Technical Values Summary

Item	Value
Failed login lockout threshold	5 attempts within 15 minutes
Account lockout duration	30 minutes
Password reset link validity	30 minutes, single use
Minimum password length	10 characters

Item	Value
TOTP code validity window	30 seconds
Backup codes issued per account	10
Website session expiration (inactivity)	90 days
Mobile app session expiration (inactivity)	180 days
API access token validity	1 hour
API refresh token validity	90 days of inactivity
Default API rate limit	100 requests/minute
Business Integration API rate limit	1,000 requests/minute
Webhook retry attempts	5, over ~24 hours

7.6 How This Document Should Be Used for RAG

As with the Billing Knowledge Base, this document is designed to be chunked by heading boundary (chapter, section, or sub-section), with quick-reference tables treated as row-level or table-level chunks so that a specific symptom or error code can be retrieved directly. The Technical Support Agent should be given Chapters 1 through 4 as primary context for customer-facing issues, Chapter 5 for developer-facing questions, and Chapter 6 whenever a customer asks about compatibility before purchasing or setting up a device.

End of Appendix and end of the TechMart Electronics Technical Support Knowledge Base document. This document is fictional and created for educational use in a Retrieval-Augmented Generation (RAG) capstone project.